EXEPRIMENT-16

QUESTION:

Write the Python Program to Implement Feed forward Neural Network

AIM:

To implement a simple Artificial Neural Network (ANN) using Python to learn the XOR logic gate by training the network with input patterns and their expected outputs.

ALGORITHM:

1. Start the program and import the required libraries.
2. Define the input patterns X and expected outputs Y for the XOR problem.
3. Initialize the weights, biases, learning rate, and number of epochs.
4. Define the sigmoid activation function.
5. Perform forward propagation to calculate the hidden-layer and output-layer values.
6. Calculate the error between the predicted output and the expected output.
7. Update the weights and biases using backpropagation.
8. Repeat the training process for 10,000 epochs to minimize the error.
9. Display the final predicted outputs for all input combinations.
10. Stop the program.

CODE:

lab16.py - C:/Users/ELCOT/AppData/Local/Programs/Python/Python314/lab16.py (3.14.6)

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```
import math
import random

# Input and expected output (XOR problem)
X = [[0, 0], [0, 1], [1, 0], [1, 1]]
y = [0, 1, 1, 0]

def sigmoid(x):
    return 1 / (1 + math.exp(-x))

random.seed(1)
hidden_size = 4
learning_rate = 0.5
epochs = 10000

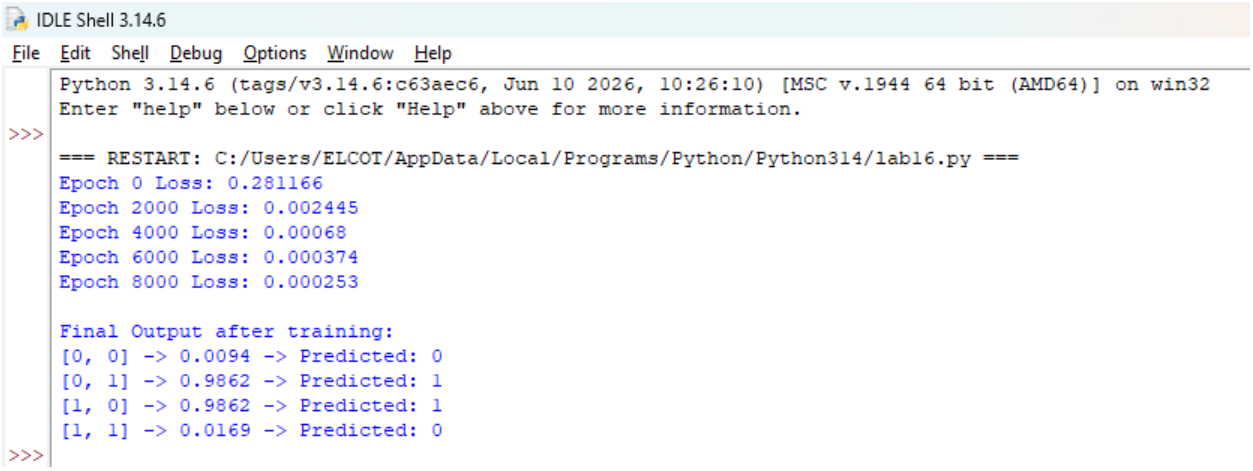
# Weights and biases
W1 = [[random.uniform(-1, 1) for _ in range(hidden_size)] for _ in range(2)]
b1 = [0.0] * hidden_size
W2 = [random.uniform(-1, 1) for _ in range(hidden_size)]
b2 = 0.0

def forward(x):
    hidden = []
    for j in range(hidden_size):
        total = x[0] * W1[0][j] + x[1] * W1[1][j] + b1[j]
        hidden.append(sigmoid(total))
    out_total = b2
    for j in range(hidden_size):
        out_total += hidden[j] * W2[j]
    return hidden, sigmoid(out_total)

for epoch in range(epochs):
    total_loss = 0
    for i in range(len(X)):
        x = X[i]
        # Forward propagation
        hidden, output = forward(x)

        # Error
        error = y[i] - output
        total_loss += error ** 2
```

OUTPUT:



RESULT:

Thus, the Artificial Neural Network was successfully implemented in Python and trained to learn the XOR logic gate. The predicted outputs were approximately 0 for inputs (0,0) and (1,1), and 1 for inputs (0,1) and (1,0), matching the expected XOR outputs